

CosmoQuest - Space Explorer Website

Frontend Web App PRD

1. Project Overview

CosmoQuest is a frontend-only space explorer website where users can explore planets, astronauts, space missions, galaxies and space facts through an animated interface.

The website should be visually animated, educational, simple to use and fully responsive. It should feel like a smooth frontend portfolio project, but it should not require backend development.

The project must use static data only. No backend, database, login, signup, admin panel, payment gateway, CMS or real API integration is required.

2. Product Goal

The goal is to build a simple but animated frontend website that makes space exploration easy and interesting.

The website should allow users to:

Goal	Requirement
Explore planets	View all 8 planets with useful details
Learn about astronauts	Browse famous astronaut cards
Discover missions	View famous space mission cards
Explore galaxies	Browse galaxy cards
Read facts	See short and simple space facts
Interact with content	Use search, filters and detail modal
Enjoy visual experience	See smooth animations and space-themed UI

3. Scope of Work

The freelancer must create a complete frontend-only website named **CosmoQuest-Space Explorer Website**.

The website must include:

- Hero section
- Planets explorer
- Astronauts section
- Space missions section
- Galaxy section
- Space facts section
- Search and filter functionality
- Detail popup/modal
- Smooth animations
- Responsive layout
- README setup file

4. Recommended Tech Stack

Recommended technologies:

Area	Recommended Tool
Frontend Framework	React
Build Tool	Vite
Styling	Tailwind CSS
Animation	Framer Motion
Data	Static JavaScript data file

5. Required Website Sections

5.1 Hero Section

The hero section must include:

- Website name: **CosmoQuest**
- Tagline: **Explore the universe from your screen**
- Short intro text
- CTA button: **Start Exploring**
- Animated star background
- Floating planet or orbit-style visual
- Smooth entrance animation for title, text and button

The hero section should create a strong first impression and clearly communicate the space exploration theme.

HomePage.png reference image will be provided.

5.2 Planets Explorer

The website must include cards for all 8 planets:

- Mercury
- Venus
- Earth
- Mars
- Jupiter
- Saturn
- Uranus
- Neptune

Each planet card must include:

Field	Requirement
Planet name	Required
Image/placeholder	Required
Short description	Required
Distance from Sun	Required
Number of moons	Required
Fun fact	Required

Each planet card should have hover animation and should open a detail modal when clicked.

5.3 Astronauts Section

The website must include at least **6 astronaut cards**.

Suggested astronauts:

- Neil Armstrong
- Kalpana Chawla
- Rakesh Sharma
- Yuri Gagarin
- Sally Ride
- Sunita Williams

Each astronaut card must include name, country, famous mission or achievement, short description and image/avatar placeholder.

5.4 Space Missions Section

The website must include at least **6 space mission cards**.

Suggested missions:

- Apollo 11
- Voyager 1
- Chandrayaan-3
- Mars Rover Mission
- Hubble Space Telescope
- Artemis Mission

Each mission card must include mission name, year, space agency, mission type and short summary.

5.5 Galaxy Section

The website must include at least **4 galaxy cards**.

Suggested galaxies:

- Milky Way
- Andromeda
- Whirlpool Galaxy
- Sombrero Galaxy

Each galaxy card must include galaxy name, galaxy type, distance from Earth, short fact and visual placeholder.

5.6 Space Facts Section

The website must include at least **8 space fact cards**.

Each fact should be short, simple, interesting and easy to understand.

Example facts:

- Jupiter is the largest planet in the Solar System.
- Saturn is famous for its rings.
- One day on Venus is longer than one year on Venus.
- The Sun contains most of the Solar System's mass.

5.7 Final CTA / Footer

The final section must include:

- Closing message
- Explore again button or scroll-to-top button
- Footer text
- Company name: **StellarNova Labs**

6. Functional Requirements

The website must include:

Feature	Requirement
Search bar	User can search by name/title
Category filter	User can filter by content type
Card grid	Cards display clearly in sections or filtered view
Detail modal	Opens when card is clicked
Smooth scrolling	Page movement should feel smooth
Hover effects	Cards and buttons should feel interactive
Responsive layout	Desktop, tablet and mobile support
Static data	All content comes from frontend data file

Required filter categories:

- All
- Planets
- Astronauts
- Missions
- Galaxies
- Facts

Search and filter must work using frontend static data only.

7. Detail Modal Requirements

When a user clicks any card, a detail popup/modal should open.

The modal must include:

- Name/title
- Larger visual or placeholder
- Full description
- Key details
- Fun fact
- Close button

The modal should open and close smoothly with animation. It should work properly on mobile screens.

8. Animation Requirements

The website should include simple, smooth, lightweight animations.

Required animations:

Area	Animation Requirement
Hero	Animated star background
Hero visual	Floating planet or orbit-style animation
Sections	Smooth reveal while scrolling
Cards	Hover animation
Modal	Smooth open and close animation
Buttons	Hover animation

Space elements	Subtle glow effect
Search/filter	Smooth transition when results change

Animations must not slow down the website or make mobile usage difficult.

9. Design Requirements

The website should feel:

- Dark space themed
- Smooth
- Educational
- Animated

Recommended colour palette:

Colour Name	Hex
Space Black	#050816
Deep Navy	#0B1026
Cosmic Purple	#6C63FF
Star White	#F8FAFC
Neon Blue	#38BDF8
Soft Grey	#94A3B8

Recommended fonts:

Usage	Font
Headings	Space Grotesk or Sora
Body	Inter or Plus Jakarta Sans

Avoid basic plain layout, too much text, messy colours, heavy 3D effects, slow animations and backend features.

10. Data Requirements

Create static data inside:

```
src/data/spaceData.js
```

The data should include:

Data Type	Minimum Count
Planets	8
Astronauts	6
Missions	6
Galaxies	4
Space Facts	8

Each data item should include:

- id
- name/title
- category
- short description

- image or placeholder
- key details
- fun fact

No real API should be used.

11. Suggested Folder Structure

```
CosmoQuest_Space_Explorer/  
├── public/  
│   └── assets/  
├── src/  
│   ├── components/  
│   ├── data/  
│   │   └── spaceData.js  
│   ├── styles/  
│   ├── App.jsx  
│   └── main.jsx  
├── package.json  
└── README.md
```

12. Final Deliverables

Submit one complete project folder named:

```
CosmoQuest_Space_Explorer/
```

The final folder must include:

- Full frontend source code
- Static data file
- Components folder
- Assets folder
- package.json
- README.md with setup instructions

13. README Requirements

The README must include:

- Project name
- Short project description
- Tech stack used
- Setup instructions
- Run command
- Build command if available

Example setup commands:

```
npm install  
npm run dev
```

14. Acceptance Criteria

The project will be accepted if:

- Website runs locally.
- Hero section is complete.
- Planet section includes 8 planets.
- Astronaut section includes at least 6 astronauts.
- Mission section includes at least 6 missions.
- Galaxy section includes at least 4 galaxies.

- Space facts section includes at least 8 facts.
- Search works.
- Filters work.
- Detail modal works.
- Animations work smoothly.
- Website is responsive.
- No backend is included.
- No console errors are present.
- README is included.
- Design feels smooth and space-themed.